EE 152: Basic Electronics (Semiconductor Basics)

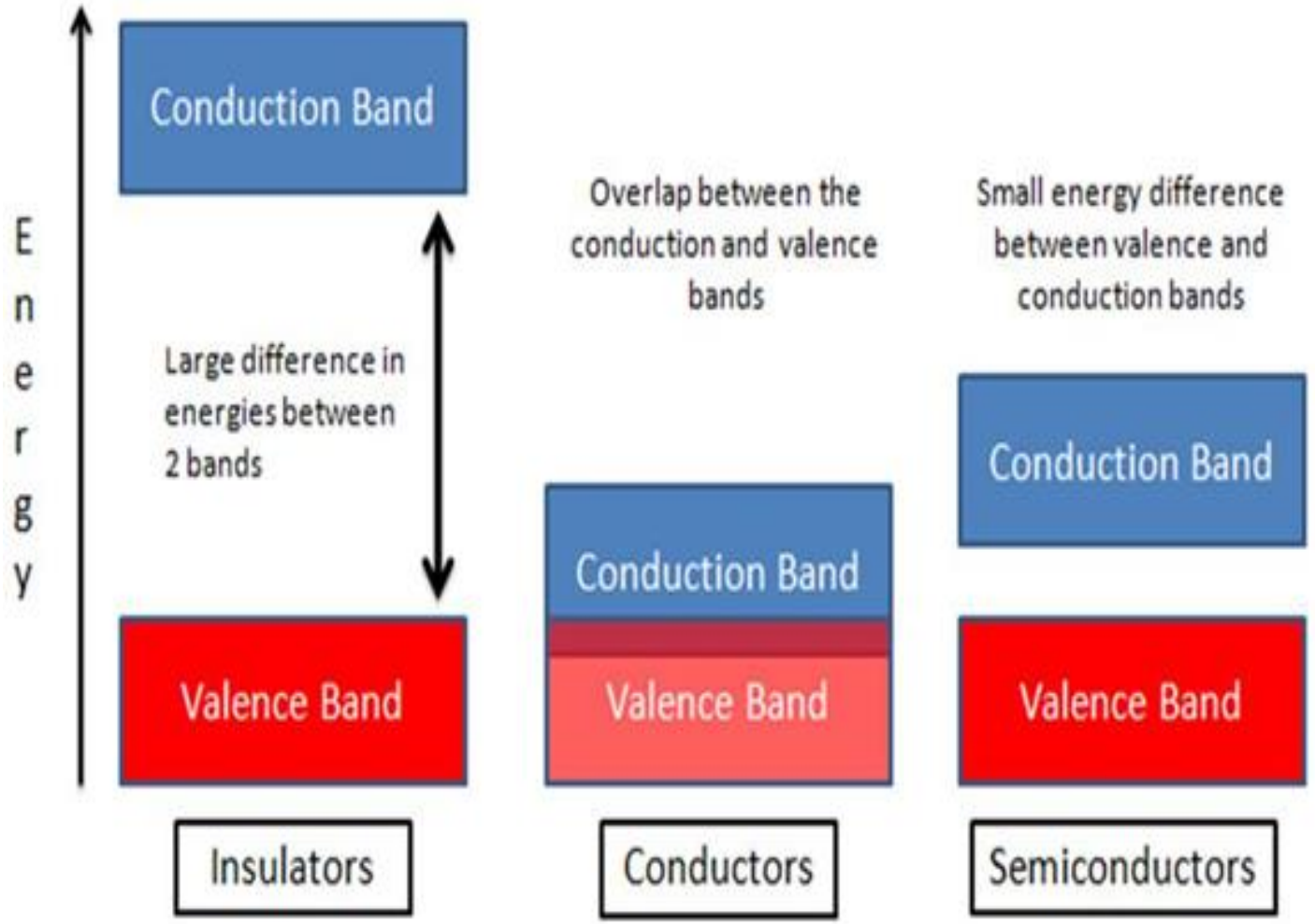
Outline

- Introduction
- Basic Semiconductor Concepts
  - Intrinsic
  - Doping
  - Extrinsic
    - N-type
    - P-type
  - Carrier movement

Electronic Materials

- The goal of electronic materials is to generate and control the flow of an electrical current.
- Electronic materials include:
  1. Conductors: have low resistance which allows electrical current flow
  2. Insulators: have high resistance which suppresses electrical current flow
  3. Semiconductors: can allow or suppress electrical current flow

Conductors, semiconductors and insulators

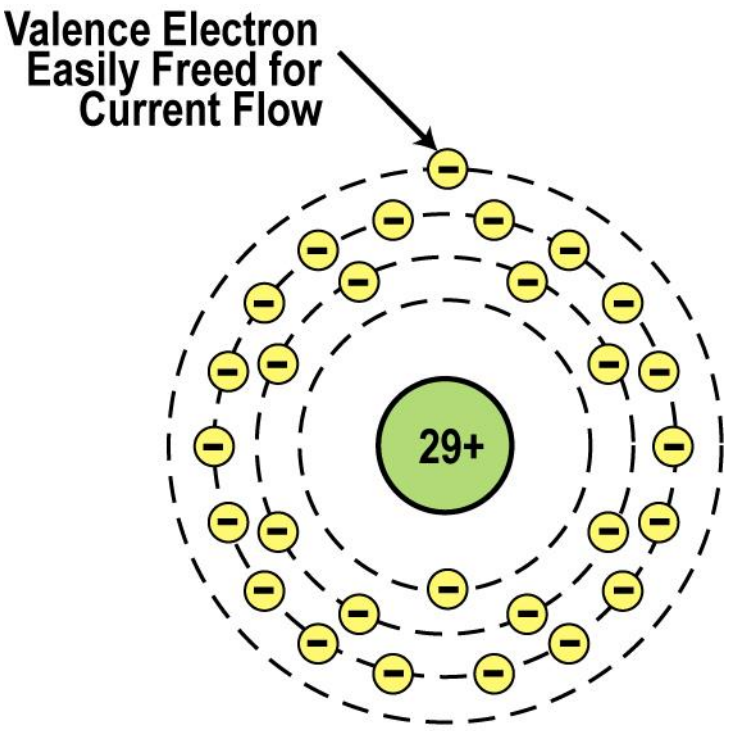


Conductors

- Good conductors have low resistance so electrons flow through them with ease.
- **Best** element conductors include:
  - Copper, silver, gold, aluminum, & nickel
- Alloys are also good conductors:
  - Brass & steel
- Good conductors can also be liquid:
  - Salt water

Conductor Atomic Structure

- The atomic structure of good conductors usually includes only one electron in their outer shell.
  - It is called a valence electron.
  - It is easily striped from the atom, producing current flow.



Copper Atom

Insulators

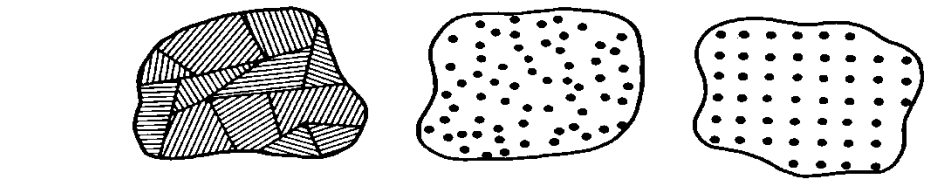
- Insulators have a high resistance so current does not flow in them.
- Good insulators include:
  - Glass, ceramic, plastics, & wood
- Most insulators are compounds of several elements.
- The atoms are tightly bound to one another so electrons are difficult to strip away for current flow.

Semiconductors

- Semiconductors are materials that essentially can be conditioned to act as good conductors, or good insulators, or any thing in between.
- Common elements such as **carbon**, **silicon**, and **germanium** are semiconductors.
- Silicon is the best and most widely used semiconductor.

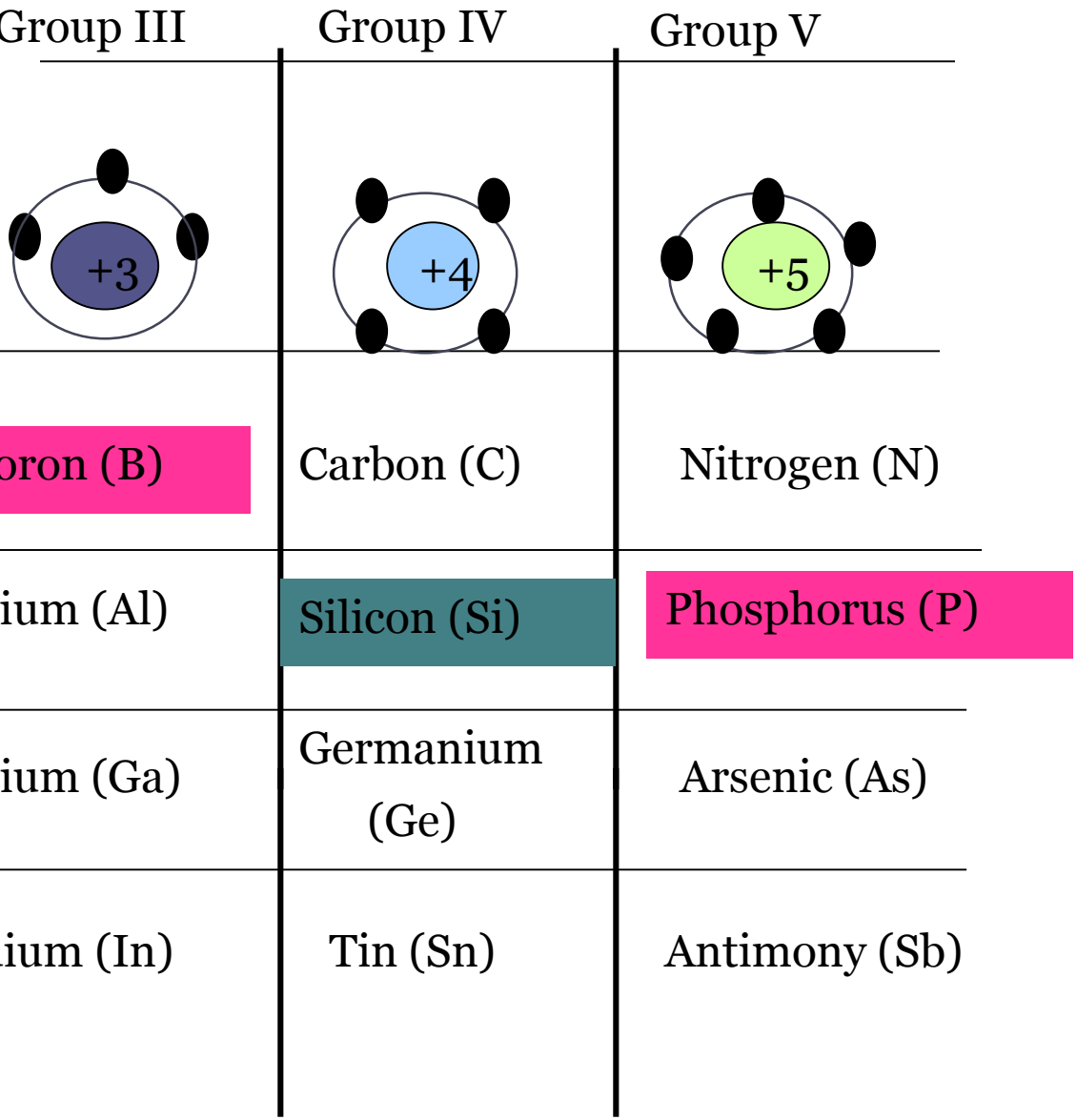
What is a Semiconductor?

- A semiconductor is a material with conducting properties between those of a good insulator (e.g. glass) and a good conductor (e.g. copper).
- The most commonly used semiconductor is silicon.
- Low resistivity => “conductor”
- High resistivity => “insulator”
- Intermediate resistivity => “semiconductor”
  - conductivity lies between that of conductors and insulators
  - generally crystalline in structure for IC devices
    - In recent years, however, non-crystalline semiconductors have become commercially very important

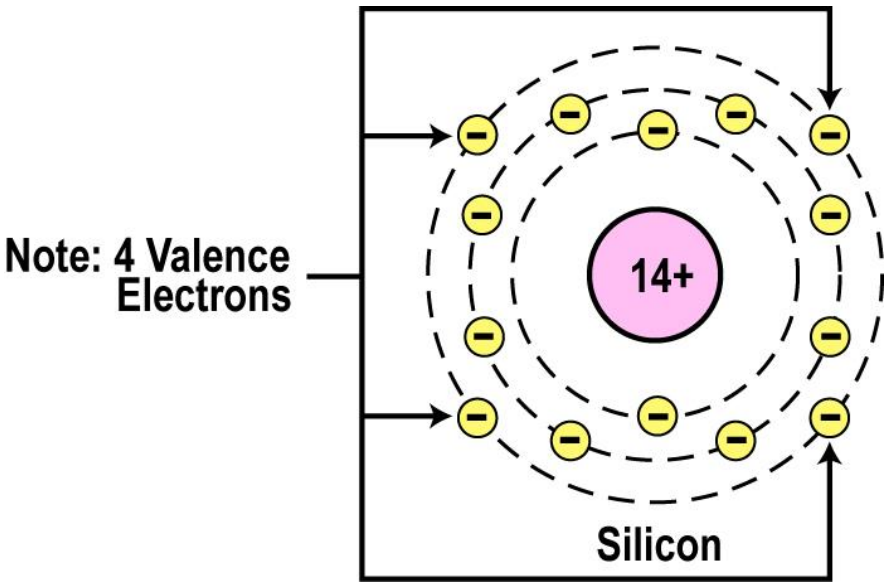


Semiconductor Elements in the Periodic Table

Group IV constitutes semiconductor elements



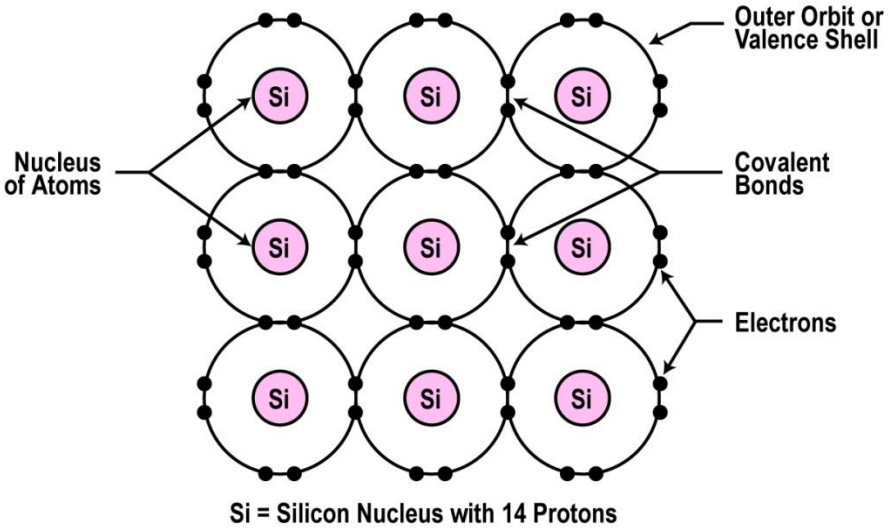
Semiconductor Valence Orbit



- The main characteristic of a semiconductor element is that it has four electrons in its outer or valence orbit.

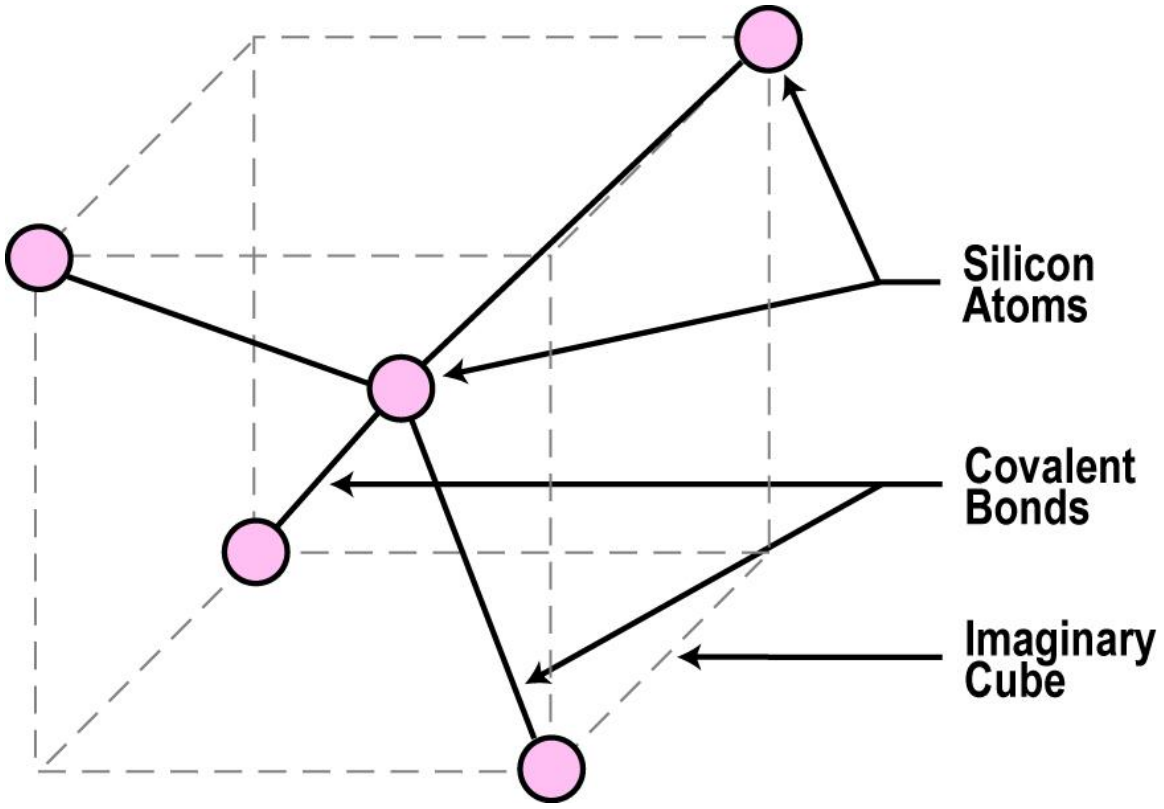
Crystal Lattice Structure

- The unique capability of semiconductor atoms is their ability to link together to form a physical structure called a crystal lattice.
- The atoms link together with one another sharing their outer electrons.
- These links are called <u>covalent bonds</u>.



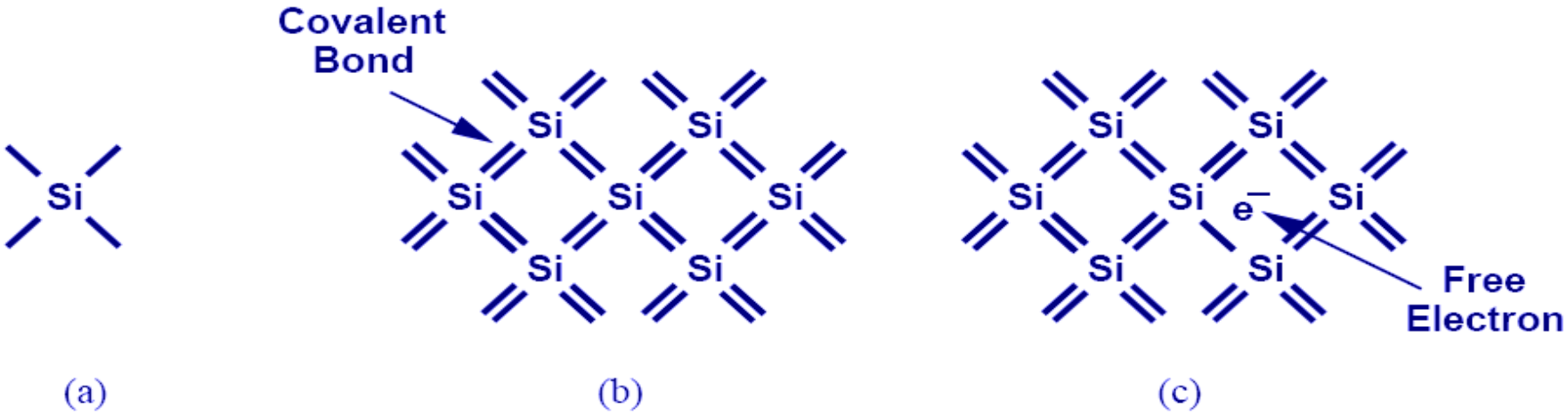
2D Crystal Lattice Structure

3D Crystal Lattice Structure



Silicon

- Atomic density: 5×10^{22} atoms/cm^3
- Each silicon atom has an outer shell with four valence electrons and four vacancies (It is a *tetravalent* element).
- In *intrinsic* (pure) silicon, atoms join together by forming *covalent bonds*. Each atom shares its valence electrons with each of four adjacent neighbours effectively filling its outer shell.
- When temperature goes up, electrons can become free to move about the Si lattice.



Electronic Properties of Si

- **Silicon is a semiconductor material.**
  - Pure Si has a relatively high electrical resistivity at room temperature.

- **There are 2 types of mobile charge-carriers in Si:**
  - ***Conduction electrons*** are negatively charged;
  - ***Holes*** are positively charged.

- **The concentration (#/cm^3) of conduction electrons & holes in a semiconductor can be modulated in several ways:**
  1. **by adding special impurity atoms (*dopants*)**
  2. **by applying an electric field**
  3. **by changing the temperature**
  4. **by irradiation**

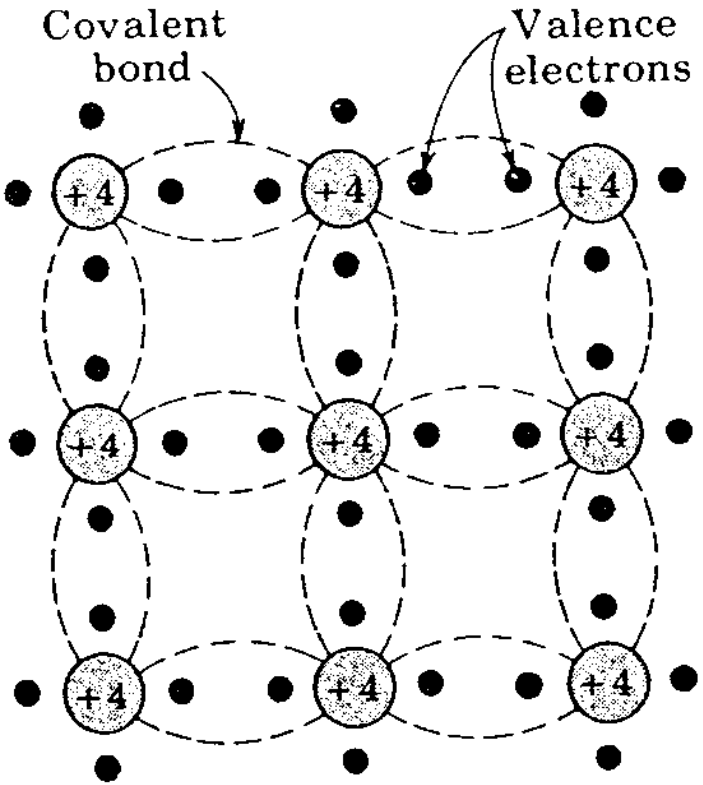
Classification of Semiconductors

- Semiconductors are classified into two types:

a) Intrinsic semiconductors

b) Extrinsic semiconductors

Intrinsic Semiconductors



Intrinsic Semiconductors

- A semiconductor is said to be intrinsic if it contains no impurities and no crystalline defects.
- The structure has zero overall charge
- At absolute zero, every electron is at the lowest energy state. Electrons occupy all of the electronic states in the valence band, and the states in the conduction band are empty.
- The intrinsic semiconductor thus behaves as a perfect insulator at absolute zero temperature.
- However, the covalent bonds can be broken if sufficient energy is applied to the semiconductor.

Intrinsic Semiconductors

- At room or higher temperatures, electrons acquire thermal energy, which is transferred to them from the crystal lattice.
- The atoms in the crystal lattice vibrate, and these vibrations can be transmitted through the crystal as acoustic waves called *phonons*.
- Phonons can excite electrons from the valence band (VB) to the conduction band (CB).
- Electrons leaving the VB to CB creates a vacancy called *hole* in the valence band. A hole is a vacancy in a covalent bond with a unit positive charge associated with it.
- The electron is now free, and there is a free hole in the valence band.
- Every electron that escapes to conduction band creates a corresponding hole in the valence band, creating an electron-hole pair.
- When electron-hole pair is created by absorption of phonons, we call the process *thermal generation*.
- If photons provide the energy, the process is termed *optical generation*.

Intrinsic Semiconductors

- The free electrons in the semiconductor can fall into holes in a process known as *recombination.*
- Energy is given up by the electrons in this process
- An intrinsic semiconductor is said to be in *thermal equilibrium* at any given temperature when the recombination rate is equal to the ionization rate
- For an intrinsic semiconductor at thermal equilibrium, the equilibrium concentration n_o of electrons in the conduction band is the same as the equilibrium concentration p_o of holes in the valence band:

$$n_o = p_o = n_i$$

- Where n_i is the equilibrium carrier concentration in an intrinsic semiconductor

Law of mass-action

- For an intrinsic semiconductor

$$n = p = n_i$$

n = concentration of electrons per unit volume

p = concentration of holes per unit volume

- This is due to fact that when an electron makes a transition to the conduction band, it leaves a hole behind in valence band .
- Thus the number of holes and electrons are equal
- Hence

$$np = {n_i}^2$$

- This equation is called as mass-action law
- This equation is valid for extrinsic as well

Carrier Concentration in Intrinsic Si

- The intrinsic carrier concentration n_i depends on;
  - the semiconductor material, and
  - the temperature
- For silicon at 300 K, n_i has a value of $\mathbf{1.5\ x\ 10^{10}\ cm^{-3}}$

Doping

- To increase the conductivity of an intrinsic semiconductor, one can introduce an impurity in a process known as doping
- These impurities are atoms from column III or V of the periodic table. They are referred to as dopants

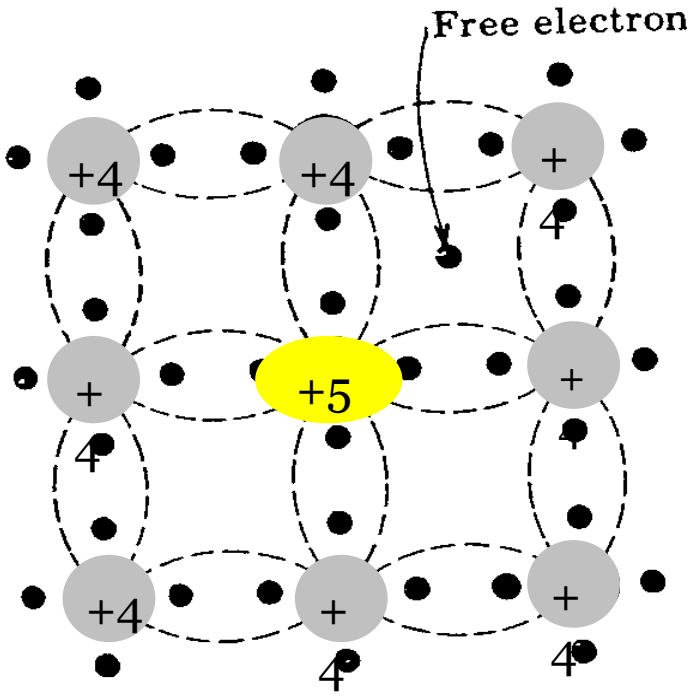
Extrinsic Semiconductors

- Intrinsic conduction is very small.
- Conductivity levels can be raised and controlled by *doping* with minute levels of impurity atoms to give *extrinsic* or *doped* semiconductors.
- Extrinsic semiconductors may be further divided into either n-type or p-type

N-type Semiconductors

- This is formed by adding a pentavalent atom to the intrinsic semiconductor
- These atoms with five valence electrons are known as donor atoms because they provide an extra electron to the semiconductor's crystal structure.
- The extra electron becomes a conduction electron.

N-type Semiconductors



Notation:
n = conduction electron concentration

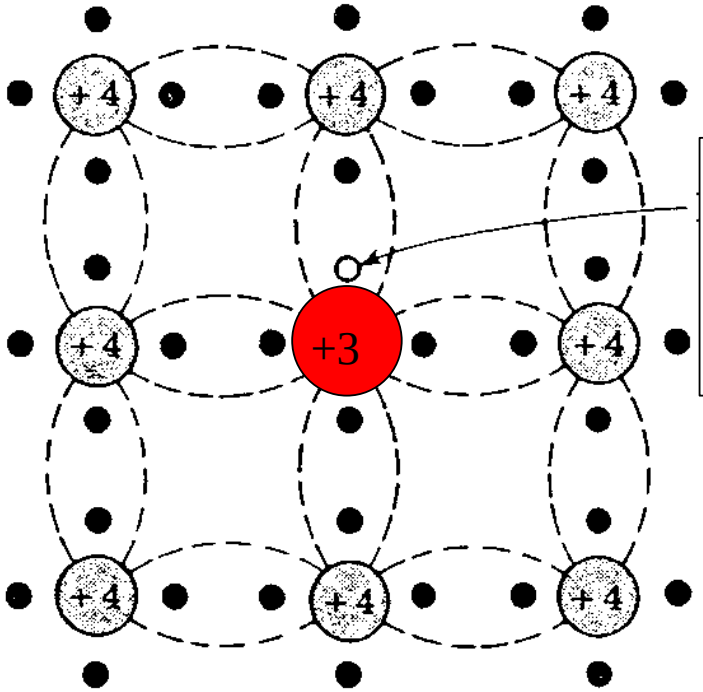
For n-type, $n \gg p$; n is the majority carrier concentration n_n

p is the minority carrier concentration p_n

P-type Semiconductors

- Here the doping atom has only three electrons in its outer shell.
- It is relatively easy for an electron from a neighbouring atom to move in, so releasing a hole at its parent atom. The freed hole is available for conduction.
- The energy needed to free the electron from its parent is usually small compared to the thermal energy so each impurity atom contributes one hole for conduction (fully ionised).

P-type Semiconductors



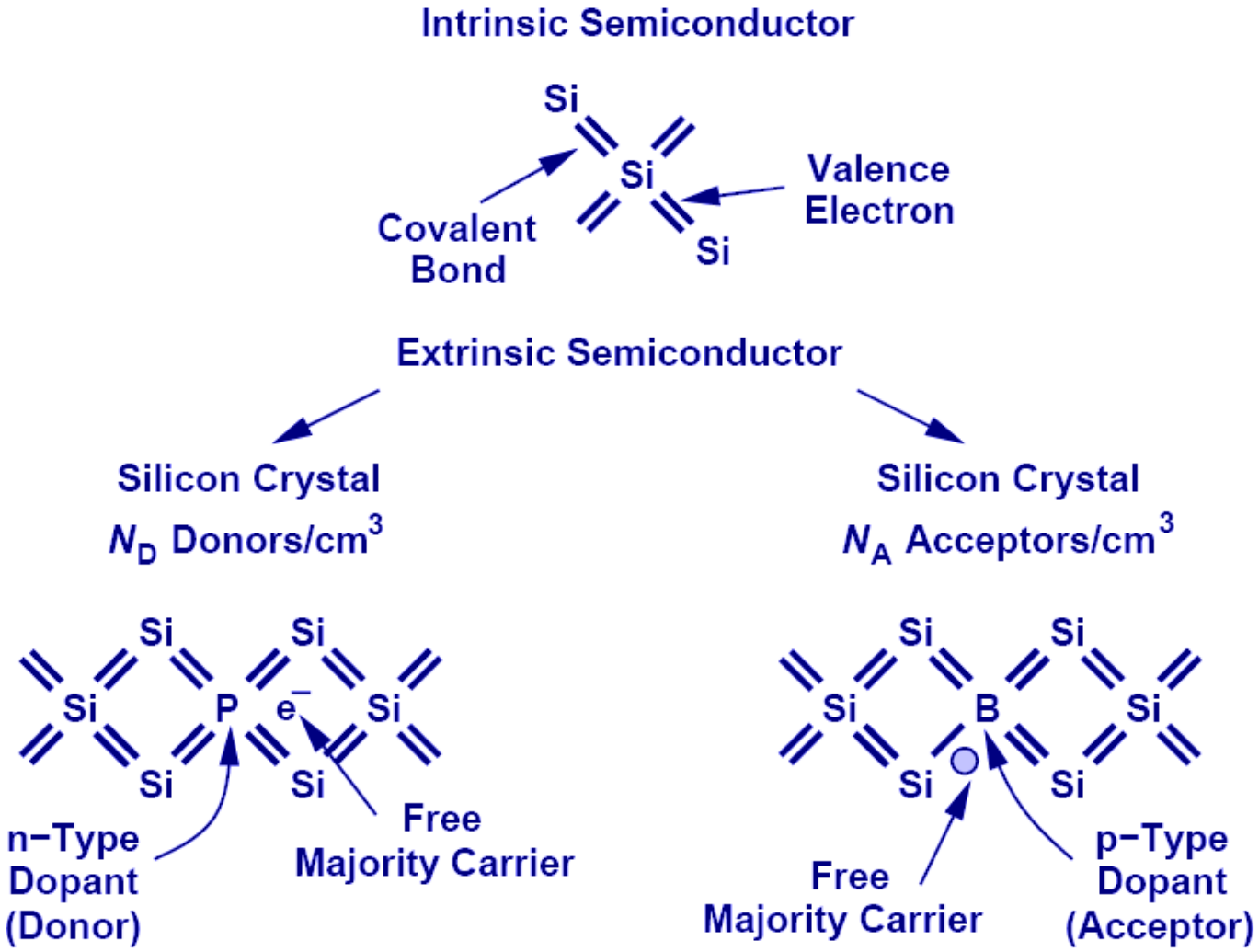
A neighbouring electron can move here. This creates a hole where the electron came from.

Notation:
p = hole concentration

For p-type, $p \gg n$; **p** is the majority carrier concentration p_p.

n is the minority carrier concentration n_p.

Summary of Charge Carriers



Electron and Hole Concentrations

- Under thermal equilibrium conditions, the product of the conduction-electron density and the hole density is ALWAYS equal to the square of n_i:

$$np = n_i^2$$

N-type material

$$n \approx N_D$$

$$p \approx \frac{n_i^2}{N_D}$$

$N_D = donor\ concentration$

P-type material

$$p \approx N_A$$

$$n \approx \frac{n_i^2}{N_A}$$

$N_A = acceptor\ concentration$

Example 1:

Calculate the hole and electron densities in a piece of p-type silicon that has been doped with $5\ x\ 10^{16}$ acceptor atoms per cm^3. $n_i = 1.5\ x\ 10^{10}\ cm^{-3}$ (at room temperature).

Solution:

P-type ; $p \gg n$

$n.p = {n_i}^2$ $\quad N_A = 5\ x\ 10^{16}$ $\quad p = N_A = 5\ x\ 10^{16}\ cm^{-3}$

$$n = \frac{{n_i}^2}{p} = \frac{(1.5\ x\ 10^{10}\ cm^{-3})^2}{5\ x\ 10^{16}\ cm^{-3}}$$

$$= 3.9\ x\ 10^3\ electrons\ per\ cm^3$$

NB:

$p \gg n_i$ and $n \ll n_i$ in a p-type material. The more holes you put in the less electrons you have and vice versa